

Домашнее задание

1. $\int (3 - x^2)^3 dx;$
2. $\int x(3 - x^2)^5 dx;$
3. $\int x(1 - 3x)^{25} dx;$
4. $\int \frac{\sqrt{x} - 2\sqrt[3]{x^2} + 1}{\sqrt[4]{x}};$
5. $\int \left(1 - \frac{1}{x^2}\right) \sqrt{x\sqrt{x}} dx;$
6. $\int \frac{2^x + 5^x}{10^x} dx;$
7. $\int \operatorname{tg}^2 x dx;$
8. $\int \frac{e^{3x} + 1}{e^x + 1} dx;$
9. $\int \frac{e^{2x} + 1}{e^x + 1} dx;$
10. $\int \frac{dx}{(x + 1)\sqrt{x}};$
11. $\int \frac{x dx}{(x^2 + 1)^2};$
12. $\int \frac{dx}{x\sqrt{x^2 + 1}};$
13. $\int \frac{dx}{1 - \cos x};$
14. $\int x(x + 1)(x - 2) dx;$
15. $\int (x^2 - 1)^2 dx;$
16. $\int 2x(x^2 - 1)^5 dx;$
17. $\int \frac{x^2 - x + 1}{\sqrt{x}} dx;$
18. $\int x^2 \sqrt{x^3 + 1} dx;$

$$19. \int x \sqrt[4]{x^2 + 1} \, dx;$$

$$20. \int x \sqrt{x + 1} \, dx;$$

$$21. \int x^3 \sqrt{x^2 + 4} \, dx;$$

$$22. \int \frac{dx}{x^2 + 4};$$

$$23. \int \frac{dx}{x^2 + 4x + 4};$$

$$24. \int \frac{dx}{x^2 + 4x + 8};$$

$$25. \int \frac{dx}{x^2 + 4x + 3};$$

$$26. \int \frac{dx}{x^2 - 9};$$

$$27. \int \frac{2x - 3}{x^2 - 9} \, dx;$$

$$28. \int \frac{x \, dx}{4x^2 - x^4};$$

$$29. \int \frac{3x - 2}{2 - 3x + 5x^2} \, dx;$$

$$30. \int \frac{2x - 1}{x^2 + 3x + 1} \, dx;$$

$$31. \int \frac{dx}{\sqrt{3 + 4x - 4x^2}};$$

$$32. \int \frac{(2x + 1) \, dx}{\sqrt{3 + 4x - 4x^2}};$$

$$33. \int \frac{dx}{\sqrt{4x^2 + 4x + 8}};$$

$$34. \int \frac{(3x + 5) \, dx}{\sqrt{4x^2 + 4x + 3}};$$

$$35. \int \frac{x \, dx}{(1 - x^2)^2};$$

$$36. \int \left(\frac{x}{x^5 + 2} \right)^4 \, dx;$$

$$37. \int \frac{x \, dx}{(1 - x)^{12}};$$

38. $\int \frac{x^5}{x+1} dx;$
39. $\int \frac{3x^2-1}{x^3-x+7} dx;$
40. $\int \frac{x dx}{x^4+6x^2+5};$
41. $\int \frac{x}{\sqrt{x-1}} dx;$
42. $\int \frac{x^2 dx}{\sqrt{x-1}};$
43. $\int x^2 \sqrt{x^2+1} dx;$
44. $\int (3x+2)e^{-3x} dx;$
45. $\int x^2 e^{-x} dx;$
46. $\int \sqrt[3]{x} \ln x dx;$
47. $\int \frac{x \arcsin x}{\sqrt{1-x^2}} dx;$
48. $\int \operatorname{arctg} x dx;$
49. $\int \sqrt{x} \ln(x+1) dx;$
50. $\int \frac{x \cos x}{\sin^3 x} dx;$
51. $\int \arccos \sqrt{\frac{x}{x+1}};$
52. $\int \operatorname{arctg} \sqrt{x} dx;$
53. $\int x \ln \frac{x+1}{x-1} dx;$
54. $\int x \operatorname{arctg}^2 x dx;$
55. $\int x^5 e^{x^3} dx;$
56. $\int \cos(\ln x) dx;$

57. $\int e^{\sqrt{x}} dx;$
58. $\int \frac{x e^x}{(x+1)^2} dx;$
59. $\int \frac{x \ln x}{(1+x^2)^2} dx;$
60. $\int \frac{2^x + 3^x}{5^x} dx;$
61. $\int \sin^5 x \cos x dx;$
62. $\int \sin^2 x \cos^3 x dx;$
63. $\int \sin^4 x \cos^2 x dx;$
64. $\int \sin 3x \cos 5x dx;$
65. $\int \sin^2 2x \cos 5x dx;$
66. $\int \frac{dx}{\sin^2(2x + \pi/4)};$
67. $\int \cos^5 x \cdot \sqrt{\sin x} dx;$
68. $\int \frac{\sin^2 x}{\cos^6 x} dx \quad (\text{замена } t = \operatorname{tg} x);$
69. $\int \frac{dx}{3 \sin^2 x - 8 \sin x \cos x + 5 \cos^2 x}$
 $(\text{замена } t = \operatorname{tg} x);$
70. $\int \frac{dx}{\sin x + 2 \cos x + 3} \quad (\text{замена } t = \operatorname{tg}(x/2));$
71. $\int \operatorname{tg}^2 x dx;$
72. $\int \operatorname{tg}^3 x dx;$
73. $\int \frac{dx}{\sqrt[3]{\operatorname{tg} x}};$
74. $\int \frac{dx}{\sin(x+1) \sin(x+2)};$
75. $\int \frac{dx}{3 + 5 \operatorname{tg} x};$

$$76. \int \frac{\sin x - \cos x}{\sin x + 2 \cos x} dx;$$

$$77. \int \operatorname{tg} x \operatorname{tg}(x + 1) dx;$$

$$78. \int |x| dx;$$

$$79. \int x|x| dx;$$

$$80. \int e^{-|x|} dx;$$

$$81. \int \max\{1, x^2\} dx;$$

$$82. \int x f''(x) dx;$$

$$83. \int e^{ax} \cos bx \, dx;$$

$$84. \int e^{ax} \sin bx \, dx;$$

$$85. I_n = \int x^n e^x dx. \text{ Найдите рекуррентную формулу для } I_n;$$

$$86. A_n = \int x^n \cos \omega x \, dx, B_n = \int x^n \sin \omega x \, dx. \text{ Найдите рекуррентное соотношение для } A_n \text{ и } B_n.$$